AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2017

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

AKU-EB May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

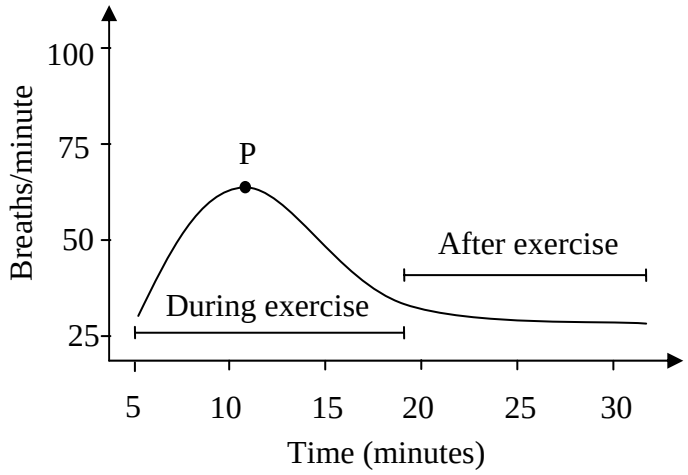
3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. The given graph shows the breathing rate of an individual during and after exercise.



i. Locate the point on the graph where lactic acid production will be the highest? (1 Mark)

ii. Why is the breathing rate of the individual high at point **P**? (1 Mark)

b. State any TWO causes of stone formation in human kidneys. (2 Marks)

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Q.2. (Total 4 Marks)

a. A biology student draws a diagram of the human heart while standing in the laboratory.

Complete the given table by identifying the parts of the student's brain against the functions performed during the activity. (2 Marks)

S. No.	Function	Part of Brain
i	It coordinates the movement of the arm for drawing and helps to keep the student in a standing position.	
ii	It helps to recall the structure of the heart that the student had studied earlier to complete the drawing.	

b. Describe the effect of vitamin A deficiency with reference to the pigment involved in human vision. (2 Marks)

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Q.3. (Total 4 Marks)

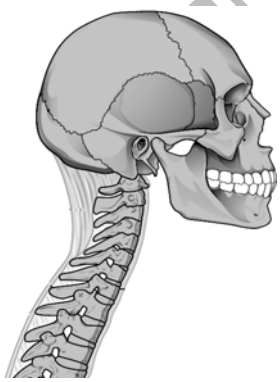
a. Fill in the given table by categorising the given parts of the human skeleton as axial or appendicular. (2 Marks)

rib cage	skull	pectoral girdle	vertebral column

Axial Skeleton	**Appendicular Skeleton**

b. The given diagram shows some parts of human skeleton.

Locate and label any ONE slightly moveable and ONE immoveable joint on the given diagram. (2 Marks)



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Q.4. (Total 4 Marks)

a. How are the following adaptive features advantageous for a wind pollinated flower?

i. Stamens are long with slender filaments. (1 Mark)

_____

_____

ii. Stigmas are large and feathery. (1 Mark)

_____

_____

b. The given table represents the data collected during a field trip to a terrestrial ecosystem.

Species	Number of Individuals
Ladybird Beetle	15
Aphids	100
Apple Tree	1

Construct pyramids of number and biomass using the given information. (2 Marks)

Pyramid of Number	Pyramid of Biomass

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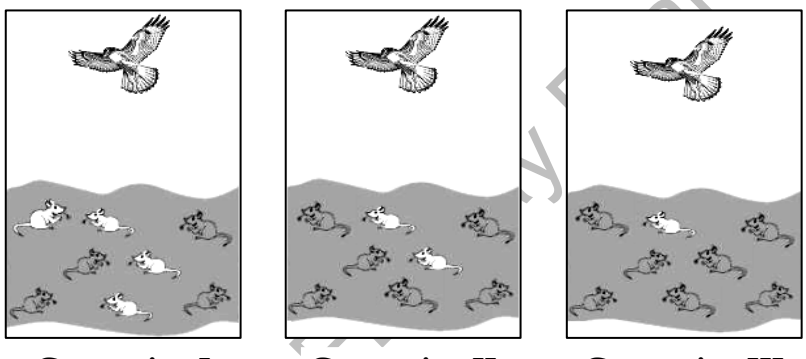
Q.5. (Total 4 Marks)

a. A template strand of DNA is shown. (2 Marks)



Draw the complementary strand of the given template.

b. The given diagrams show the comparative population of mice of two different colours over three generations with their predator.



Generation I **Generation II** **Generation III**

Describe the phenomenon responsible for the survival of grey mice. (2 Marks)

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Q.6. (Total 5 Marks)

a. Mention an advantage of using fermenters in industries. (1 Mark)

b. State any TWO conditions for which sedatives are prescribed by doctors. (2 Marks)

c. State the contribution of the following scientists in the field of pharmacology.

i. Alexander Fleming (1 Mark)

ii. Joseph Lister (1 Mark)

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Q.7. (Total 7 Marks)

EITHER

a.

i. Mention any ONE function of hair present in the nasal cavity. (1 Mark)

ii. Describe the structure and function of the following parts of the human respiratory system. (6 Marks)

I. Larynx
II. Trachea
III. Alveoli

OR

b. Pancreas acts both as endocrine and exocrine gland.

With reference to the endocrine nature of pancreas, mention

i. the names of pancreatic cells and the hormones they produce. (3 Marks)
ii. any ONE way in which each of these hormones maintain the blood glucose concentration. (2 Marks)
iii. any ONE effect of hypo secretion of each hormone on the body. (2 Marks)

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Q.8. (Total 8 Marks)

EITHER

a. Two types of reproduction are shown in the given diagrams.

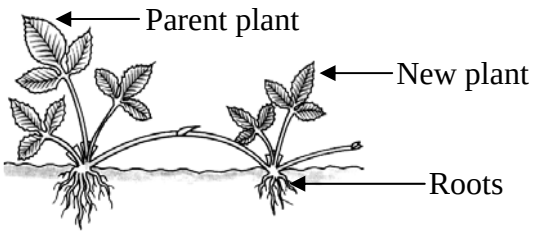


Diagram I

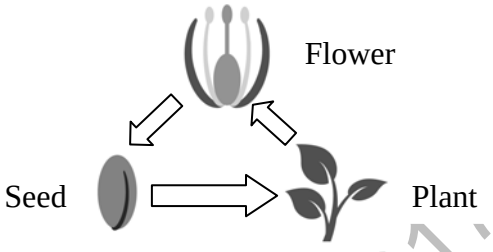


Diagram II

i. Name both types of reproduction. (2 Marks)

ii. Describe any FOUR advantages and TWO disadvantages of the type of reproduction shown in **Diagram I**. (6 Marks)

OR

b. A man with blood group A marries a woman with blood group B. Two of their children have blood group O, one has blood group B and one has blood group AB.

i. Define co-dominance. (1 Mark)

ii. Referring to the given statement, describe the genotype, antigen produced and the relationship between alleles. (7 Marks)

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Please use this page for rough work

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